

Devices

- **Hub** -
 1. It doesn't have any memory.
 2. It works on Half Duplex.
 3. It is a Physical Layer device.
 4. It broadcasts the signal.
 5. It has multiple ports to connect many of devices.
 6. It is used in LAN.
- **Switch** - It has memory to store mac address and form mac table according,
 - Its layer two devices
 - It has multiple port so that we can connect many devices
 - It is use to connect single network device
 - Most commonly used for LAN connectivity.
- **Router** - it is used to connect two diff-2 LAN
 - Routers can be use in LAN but most commonly used in WAN
 - Router work in IP Address, Router is layer 3 device
- **Bridge** - Only have two ports
 - it has small memory it is work in mac address table
- **Firewall** - it is used to control inbound and outbound traffic to protect your network
 - it is work on network layer3, it works on IP and MAC address, it is work on layer7(user interface)

TCP/IP Model:-

Application layer	Application layer
Transport layer	Host to Host
Network layer	Internet layer
Data link layer	Network Access layer
Physical layer	

The protocol that make up TCP/IP define:-

- * How data is transmitted across a network
- * How data should be formatted so that network system can understand it.

Two key TCP/IP features support Decentralistion:-

1. End Node Verification:- The two endpoint of any data transfer are responsible for making sure it was successful.
 - * no centralised control system.
2. Dynamic routing:- End nodes can transfer data one multiple paths and the n/w chooes the best path for individual data transfer.

Network issues:-

1. Addressing
2. Routing
3. Name resolution
4. Flow& Error Control
5. Interoperability

MSS: Maximum Segmentation Size

MSS – 1460 bytes	
1 byte	8bits
8bit	1 byte
1024byte	1MB (Megabytes)
1024MB	1GB (Gigabytes)
1024GB	1TB (Terabytes)
1024TB	1PB (Petabytes)
1024PB	1EB (Exabytes)
1024EB	1ZB (Zettabytes)
1024ZB	1YB (Yottabytes)

Protocols

Protocols	HTTP	HTTPS	Telnet	SSH	SMTP	FTP	DHCP	TFTP	DNS
	TCP	TCP	TCP	TCP	TCP	TCP	UDP	UDP	UDP/TCP
Port Numbers	80	443	23	22	25	20/21	67/68	69	53

Default Gateway:-

- * Any L3 Device through which any LAN user communicates outside the LAN.
- * The IP assigned on to the Interface Connected to the LAN will be your default gateway IP.

Icmp(Internet control massaging protocol):-

- * Is used to verify connectivity b/w source and destination.

- * Utilities/ tools:-

ping = packet internet groper.

Traceroute = Tracing the path b/w source to destination.

Messages:- --- Query = echo request

Echo reply

User ---- Windows - 5sec time out == 4 time

Cisco - 2sec timeout === 5 time

Arp(Address resolution protocol):-

- * Arp finds mac address of a host from its know ip address.**
- * It is layer 2 protocol.**
- * Arp request is a broadcast but arp response is a unicast.**
- * The arp concept used only in IPv4.**
- * Arp is never generated for the device of other network.**

Arp is four types:-

- 1. ARP**
- 2. Proxy Arp**
- 3. Reverse Arp**
- 4. Gratuitous**

HTTP(Hyper text transfer protocol):-

- * To exchange web pages b/w client to server.**
- * Encapsulates in TCP with port no 80.**

- * It is not encrypted or secure.
- * All data is send in cleartext form.

HTTPS(Secure):-

- * Encapsulates in TCP with port no 443.
- * All data is encrypted.
- * Encryption is done for https will be done by presentaton layer.

FTP(File Transfer Protocol):-

- * It is use to transfer data b/w client to server.
- * Encapsulates in TCP with port no 20,21
- * Data is unsecured.
- * We can browser through files and folders.
- * you can not download or copy folder but only file and that to one by one.

TFTP(Trivial FTP):-

- * Encapsulates in UDP Port no 69.
- * No Authentication.
- * Data is unsecured and cannot be secured in any way.
- * In order to secure any file, it is recommended to assign a typical file name.

Mac Address: Media access control address -: MAC address is a unique identifier assigned to a network interface control (NIC) for use as a network address in communication within a network segment. Mac address global unique. MAC address size 48 bit / 8bit = 6byte. Mac address representing hexadecimal format.

Physical address		
CISCO MAC	Mobile MAC	Laptop Mac
34AB: CDFE:6949	34: AB:CD: FE:69-49	34-AB-CD-FE-69-49

48bit (Define by IANA): 24-bit organisation | 24 vendors specific.

DHCP: Dynamic host configuration protocol – It is used for the IP implementation -> Statically and dynamically

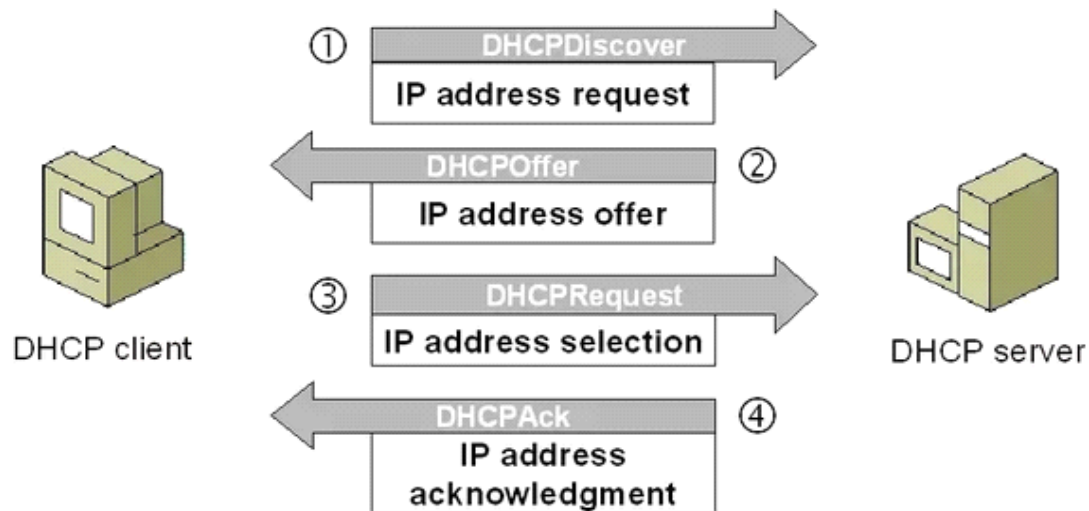
- It is automatically assigned IP address to DHCP client.
- It is based on client server model
- DHCP work at application layer
- IP address assigned is known as dynamic IP address
- DHCP IP address range is called scope.
- DHCP is dynamic boot up
- It is use UDP 67/68 Ports in transport layer (67 for server 68 is used for client)

DHCP server provide service

- IP Address
- Subnet Mask
- Default gateway
- Domain name
- DNS server address

DORA PROCESS

DORA: Discover, offer, request, Acknowledgement): it is type of message in DHCP



Discover/client (68): UDP broadcast from DHCP client to local available server

- Layer 2 Broadcast =
- Layer 3 Broadcast = 255.255.255.255

Offer/server (67): DHCP server to client in response to DHCP discover with offer of configuration parameters

(IP ADDRESS of DHCP server offer IP, MAC address of client subnet mask).

Request/client (68): Client then broadcast to the server a DHCP request message asking for the offered IP.

Acknowledgement/Server (67): server to client with configuration parameters including connect network address.

APIPA(Automatic private IP address):-

==> 169.254.0.1 ----- 169.254.255.254

==> Inbuilt feature

Advantage:-

* It works same as DHCP server.

* It stop unwanted broadcasting.

DNS (Domain name server/service/system)

- It is used to translate IP into domain names.
- It is used to 53 protocol number.
- It is work on application layer.

Telnet (Teletype Network)

- It is using for the remote access of other devices
- It is using port no 23
- All the data that is transmitted through telnet is in cleartext format
- encapsulates in TCP protocol

SSH (Secure shell)

- Almost same as telnet
- All data is encrypted and secure
- It is using port number - 22
- Encapsulates in TCP protocol
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Routing Part

Ip Routing: To identify the path from on network to another network

Types of routing:

Static Routing:

- with using next hop address
- with using exit interface
- with using exit interface and next hop address

As number (autonomous system number): Group of devices or network within a single administrator. It is a 16bit decimal number. Range = 0-65535

Dynamics routing: Routing in which routes are calculated automatically

Routers automatically exchange the *routes* b/w each other

Some packets are there for each protocol which are advertised by routing protocols

Types of dynamics routing:

IGP – Interior gateway protocol

EGP – Exterior gateway protocol

IGP - Routing protocols that share routes among some autonomous system (Exp: EIGRP, OSPF)

EGP – Routing protocol that shares routes b/w different autonomous systems (Exp: BGP)

ACL(Access control list):-

* ACL is a set of rules which will allow or deny the specific traffic moving through the router.

* It is layer 3 security which controls the flow of traffic from one router to another.

* It is also called as packet filtering firewall.

Type of ACL:- 1. Standard

2. Extended

Standard Access list

* The access-list number range is 1 - 99

* Can block a network, Host and subnet.

* Two way communication is stopped.

* All services are blocked.

* Implemented closest to the destination.

Extended access list

* The access-list number range is
100-199.

* Can block a network , host,
Subnet and service

* Implemented closest to the
source.

* One way Communication is
Stopped.

* Selected services can be Blocked.

* Filtering is done based on only source ip

address

NAT (Network Address Translation):-

Network Address Translation is a process of changing the source and destination IP addresses and ports. Address translation reduces the need for IPv4 public addresses and hides private network address ranges. This process is usually done by routers or firewalls.

Three types of address translation:-

Static NAT – Translates one private IP address to a public one. The public IP address is always the same.

Dynamic NAT – Private IP addresses are mapped to the pool of public IP addresses.

Port Address Translation (PAT) – One public IP address is used for all internal devices, but a different port is assigned to each private IP address. Also known as NAT Overload.

CDP Cisco Discovery Protocol:-

- it is a cisco proprietary protocol
- it works on layer2
- it is by default enabled on cisco devices
- it is used to discover the information about its immediate neighbor
- this information will be helpful in case of troubleshooting
- cdp timer 60sec, hold down 180sec
- multicast mac address 0100.0000.0000
- switch# show cdp neighbor details
- switch# show cdp neighbor

how to enable cdp

- switch(config)#cdp run

- switch(config)#no cdp run
- switch(config-if)# cdp enable
- switch(config-if)# no cdp enable

how to change cdp timer

- switch(config)# cdp timer 80
- switch(config)# cdp holddown 200

LLDP Link Layer Discovery Protocol:-

- it is a open standard protocol
- it is used to discover the information about its immediate neighbor
- this information will helpfull in the case of troubleshooting
- LLDP timer 30sec, LLDP holddown 120sec
- it will send its all information in TLV (type length value)

LLDP configuration

- switch(config)#lldp run
- switch(config-if)# lldp transmit
- switch(config-if)# lldp receive
- switch# show lldp

➤ switch# show lldp neighbors

➤ switch# show lldp entry

VLAN virtual local area network:-

- To break broadcast domain at layer2 we use vlan and one vlan defines one broadcast domain.
- vlan identify by a number and it is in 12bits.
- range of vlan from 0 to 4095 (normal=1to1005,extended=1006 to 4094).
- default vlan is vlan1.
- vlan database save in vlan.dat file in flash memory.

how to create vlan:-

- switch(config)#vlan 10
- switch(config-vlan)#name sales
- switch#show vlan brief

how to assign port to any vlan:-

Static vlan assignment:

- switch(config-if)# switchport access vlan 10
- switch#show vlan brief
- switch#show vlan id 10

Dynamic vlan assign:-

- VMPS Vlan Management Policy Server
- AAA Authentication, Authorization and Accounting.

Switchport:-

a switch which has capability to do layer2 address switching

- Access port- carry single vlan data
- Trunk port - carry multiple vlan data